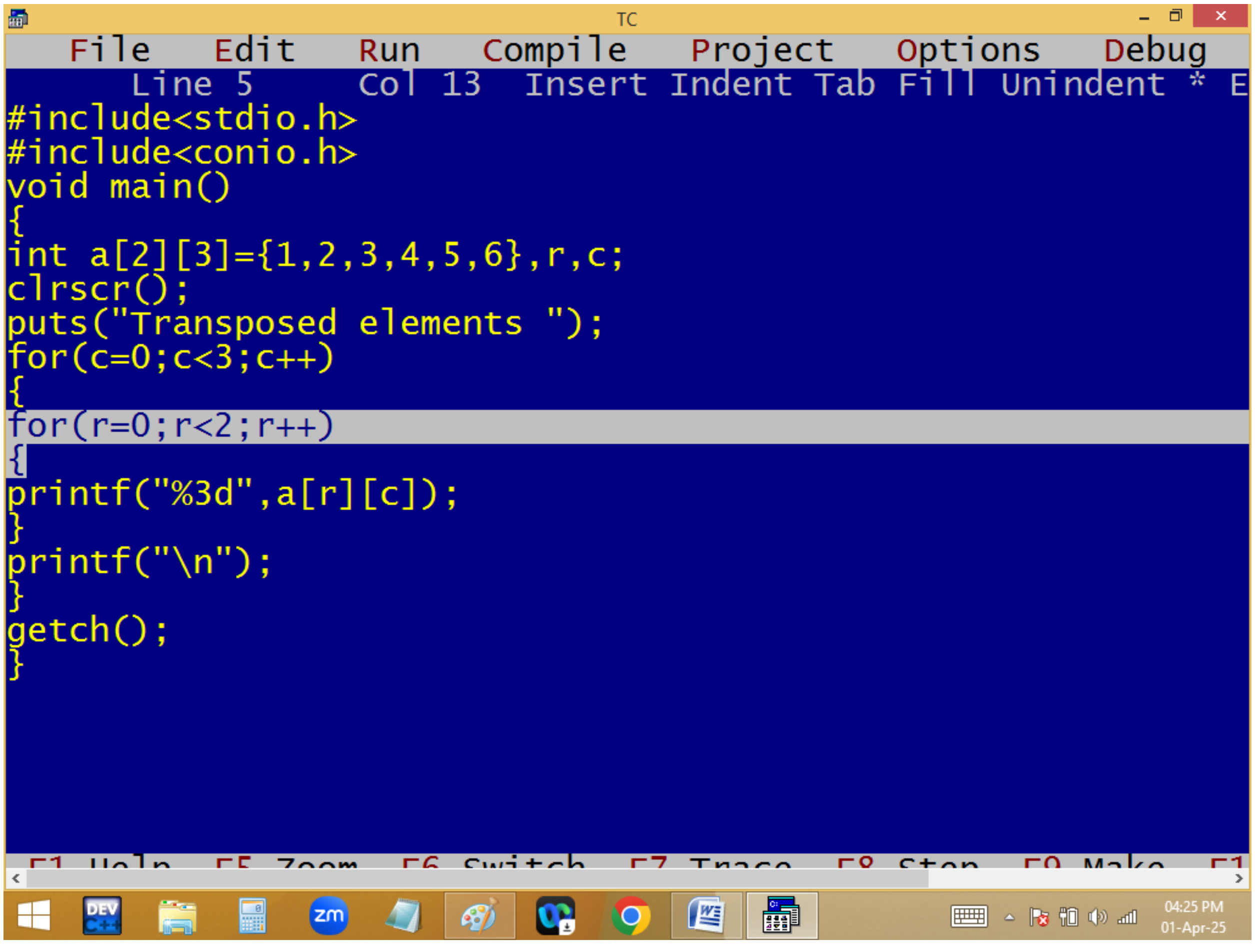
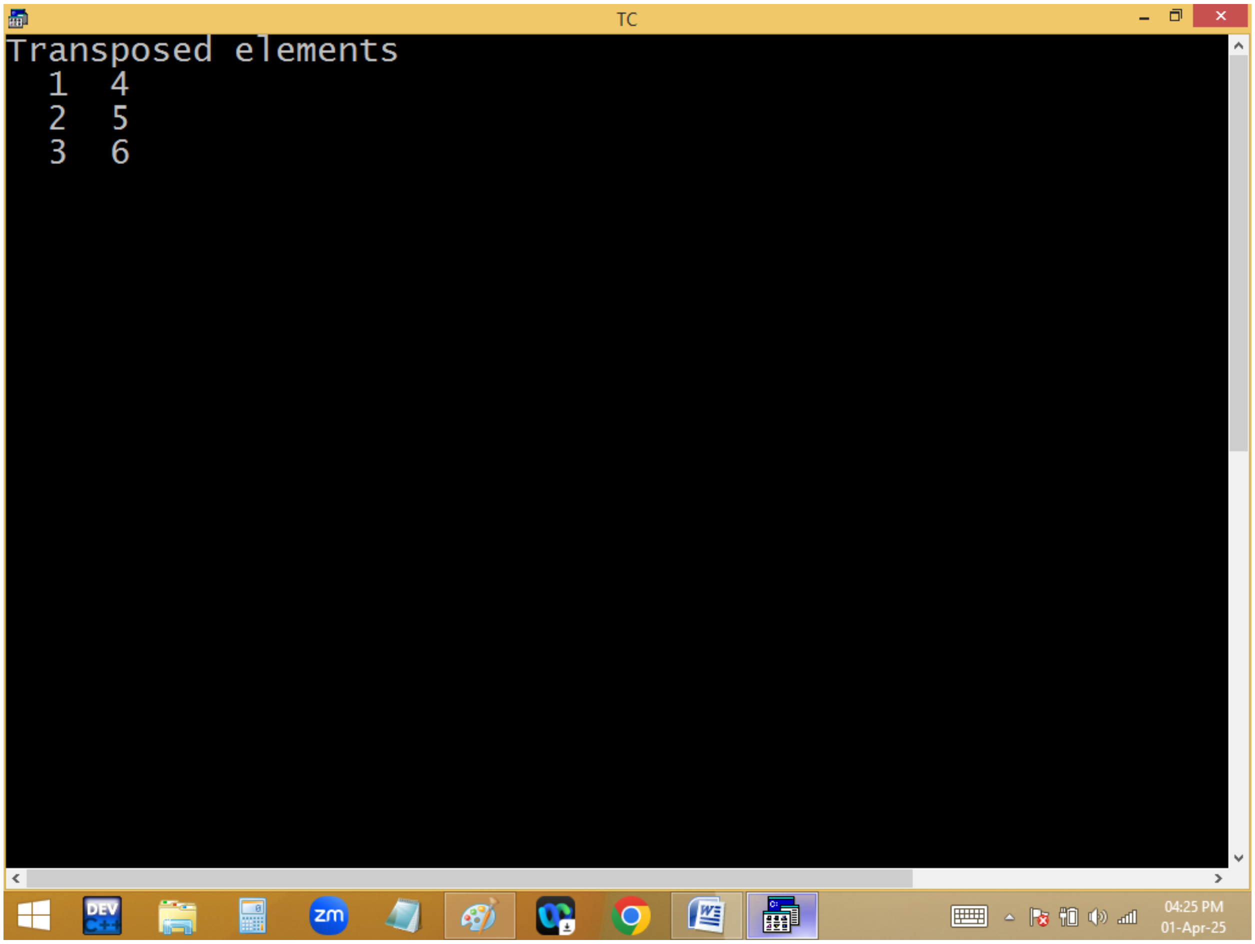
Transpose of n*n matrix:






```
for( c=0; c<3;c++)
{
for( r=0; r<2; r++ )
{
p( a[ r ][ c ] );
}
p("\n");
}
```

r	c
0 1 ~~2~~	0
0 1 ~~2~~	1
0 1 ~~2~~	2
	~~3~~

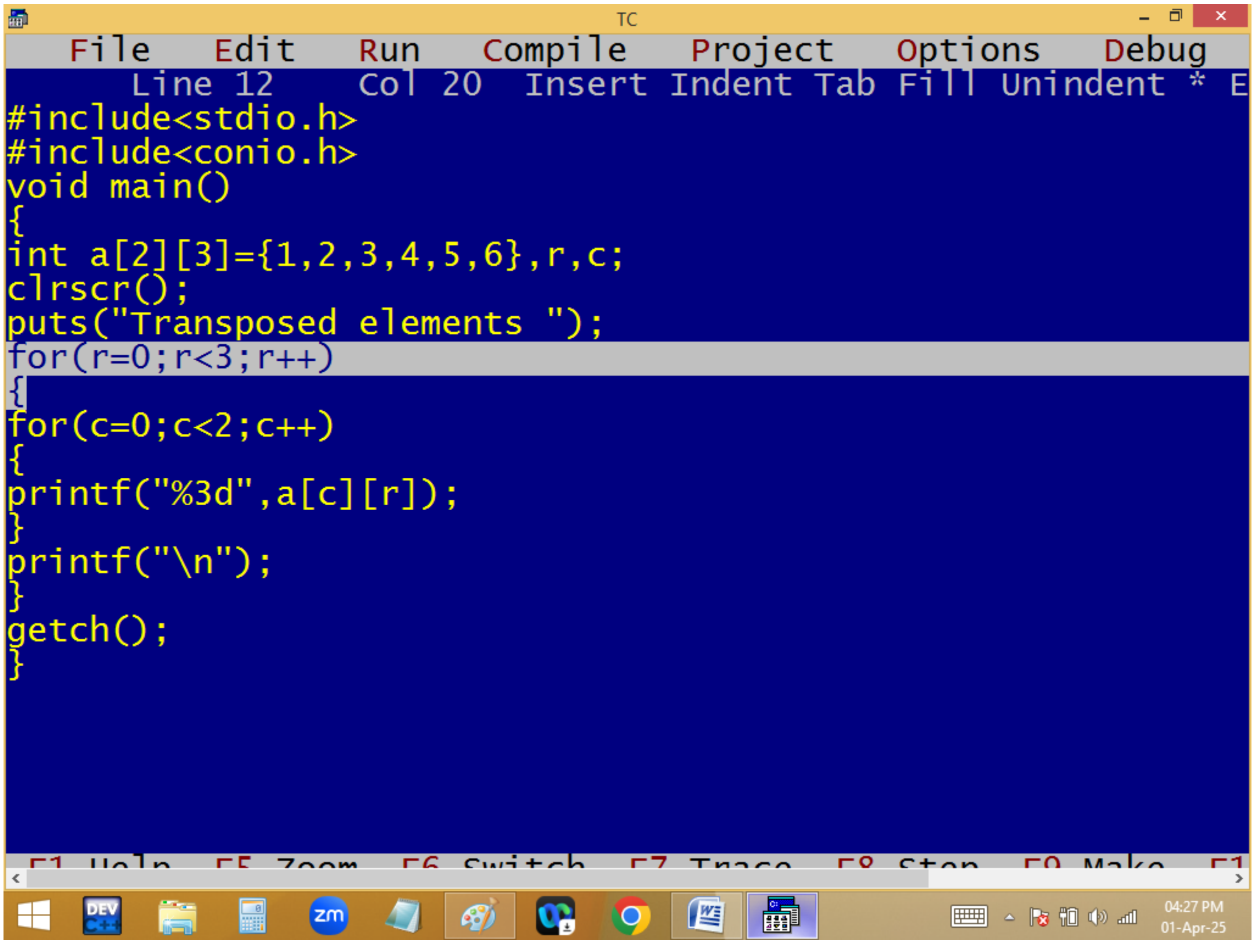
2*3

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2

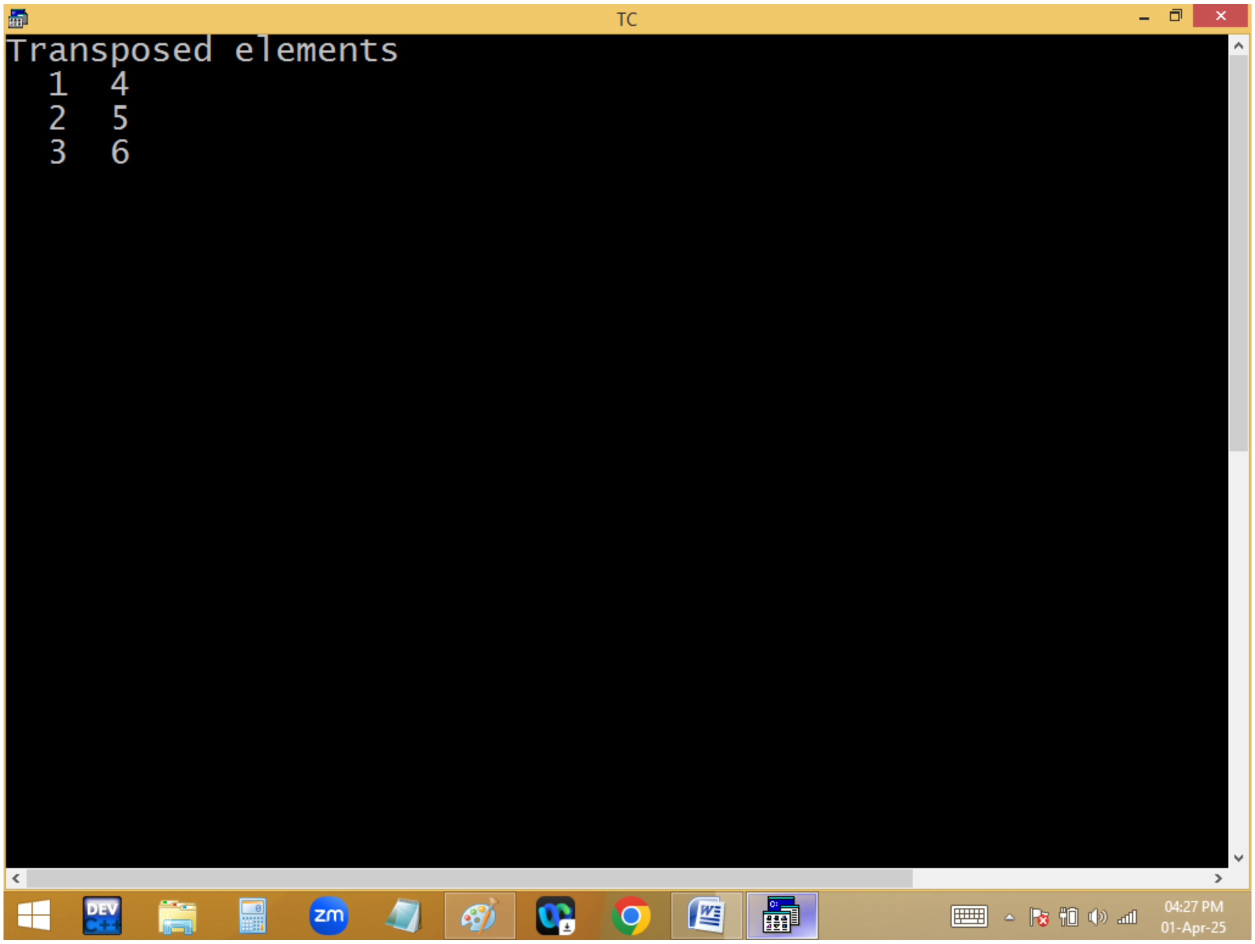
3*2

1	4
2	5
3	6

Method2:



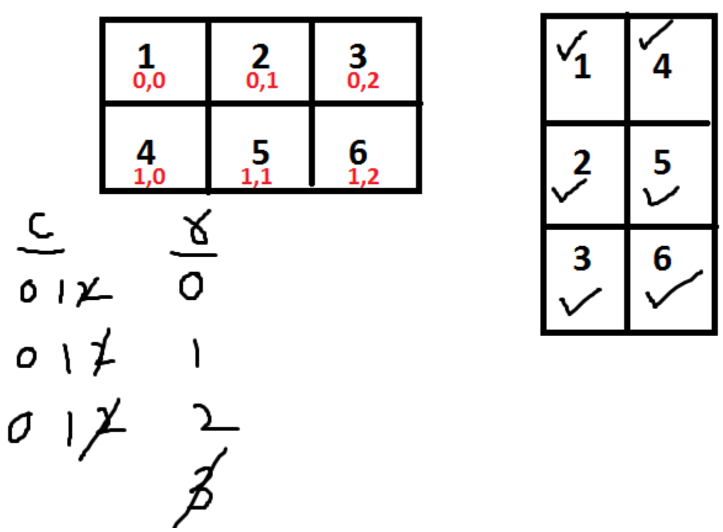

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
puts("Transposed elements ");
for(r=0;r<3;r++)
{
for(c=0;c<2;c++)
{
printf("%3d",a[c][r]);
}
printf("\n");
}
getch();
}
```




```
for( r=0; r<3; r++ )
{
for( c=0; c<2;c++)
{
p( a[ c ][ r ] );
}
p("\n");
}
```

2*3

3*2



TC

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Line 9 Col 34 Insert Indent Tab Fill Unindent * E

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Transposed elements ");
for(r=0;r<nc;r++)
{
for(c=0;c<nr;c++)
{
printf("%3d",a[c][r]);
}
printf("\n");
}
getch();
}
```

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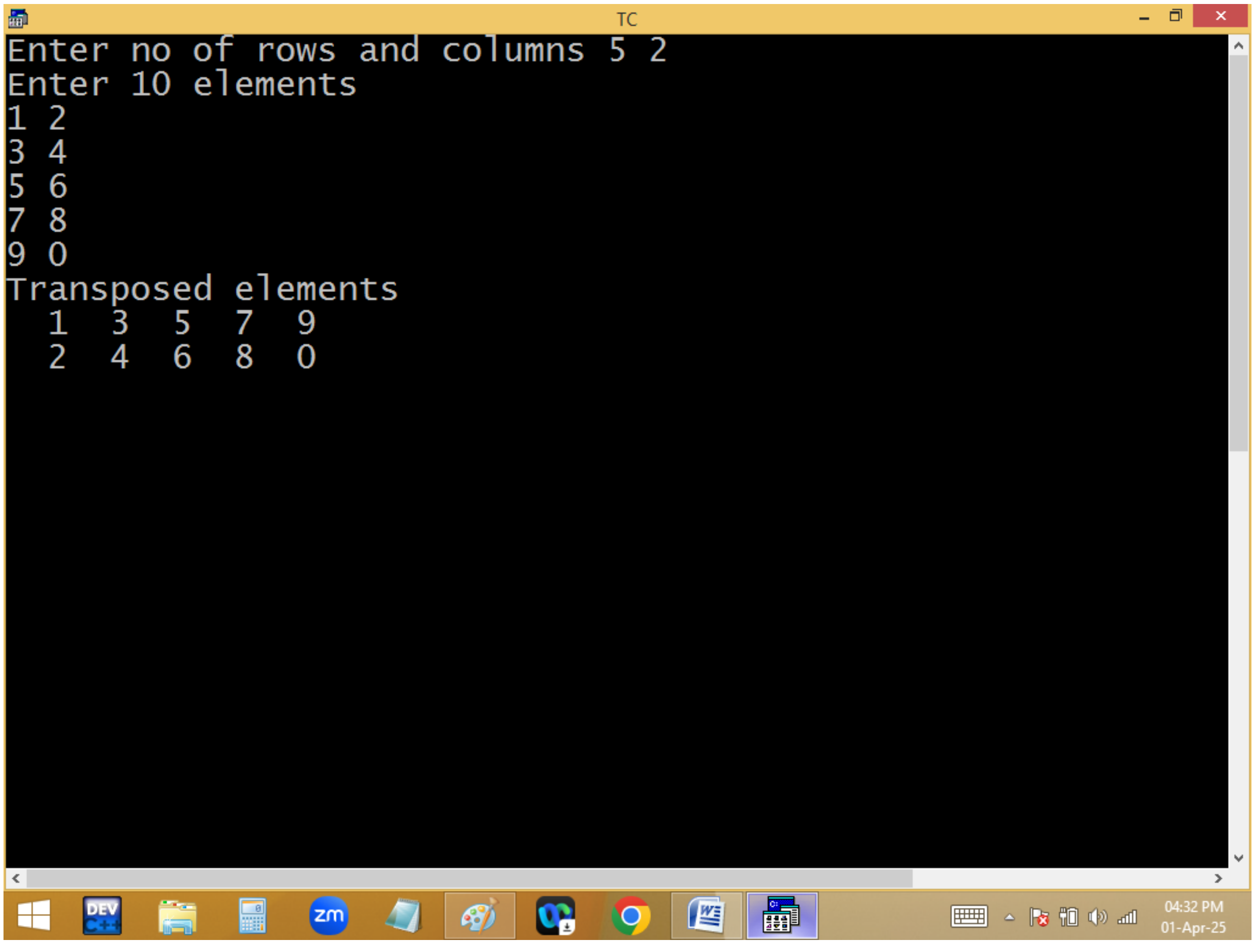
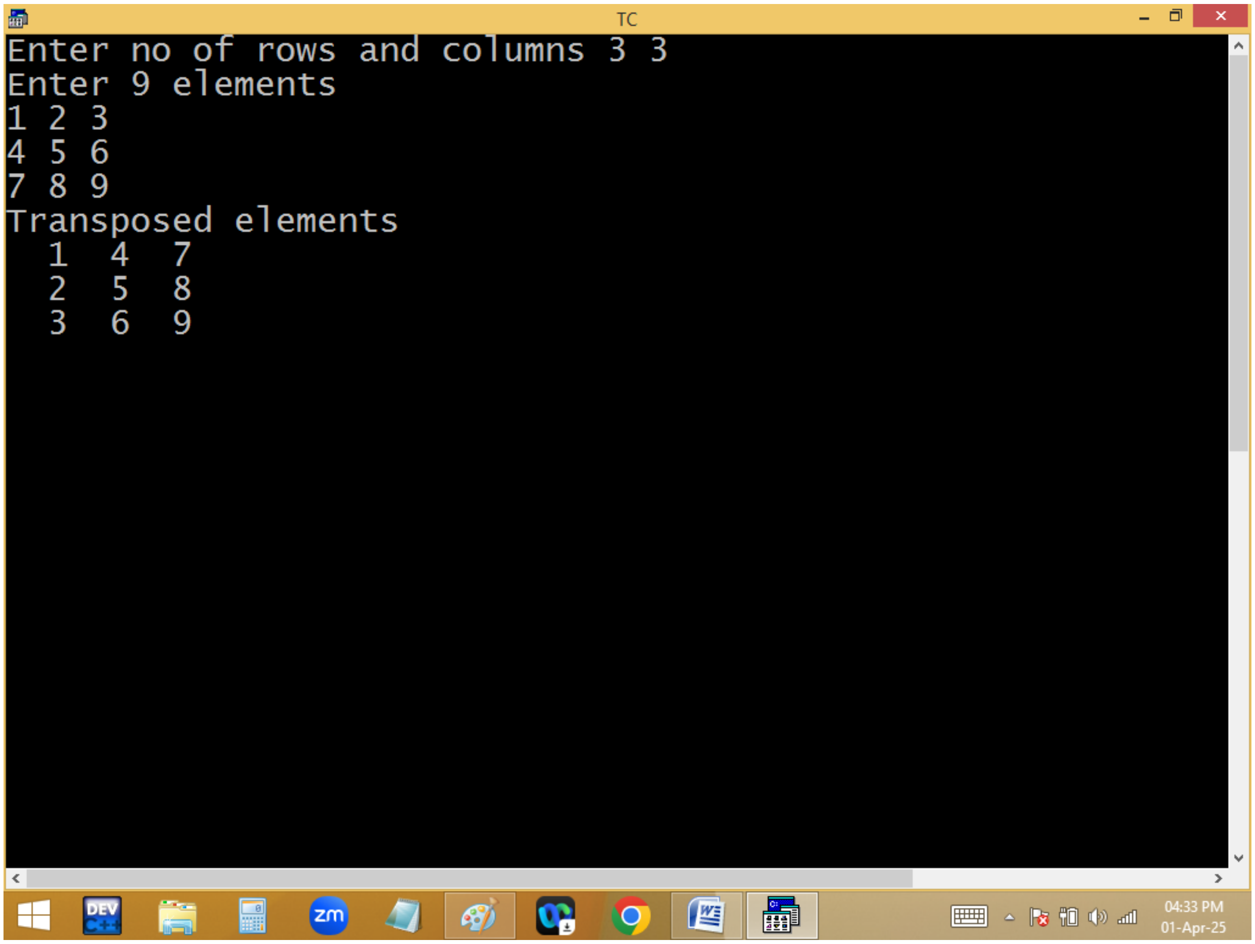
01-Apr-25


TC
Enter no of rows and columns 5 2
Enter 10 elements
1 2
3 4
5 6
7 8
9 0
Transposed elements
1 3 5 7 9
2 4 6 8 0
04:32 PM
01-Apr-25



Finding trace of n*n matrix:

TC

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Line 1 Col 9 Insert Indent Tab Fill Unindent * E

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,s=0;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
if(nr==nc)
{
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)
{scanf("%d",&a[r][c]); if(r==c)s+=a[r][c];}
printf("Trace=%d",s);
}
else printf("Rows and columns should be same");
getch();
}
```

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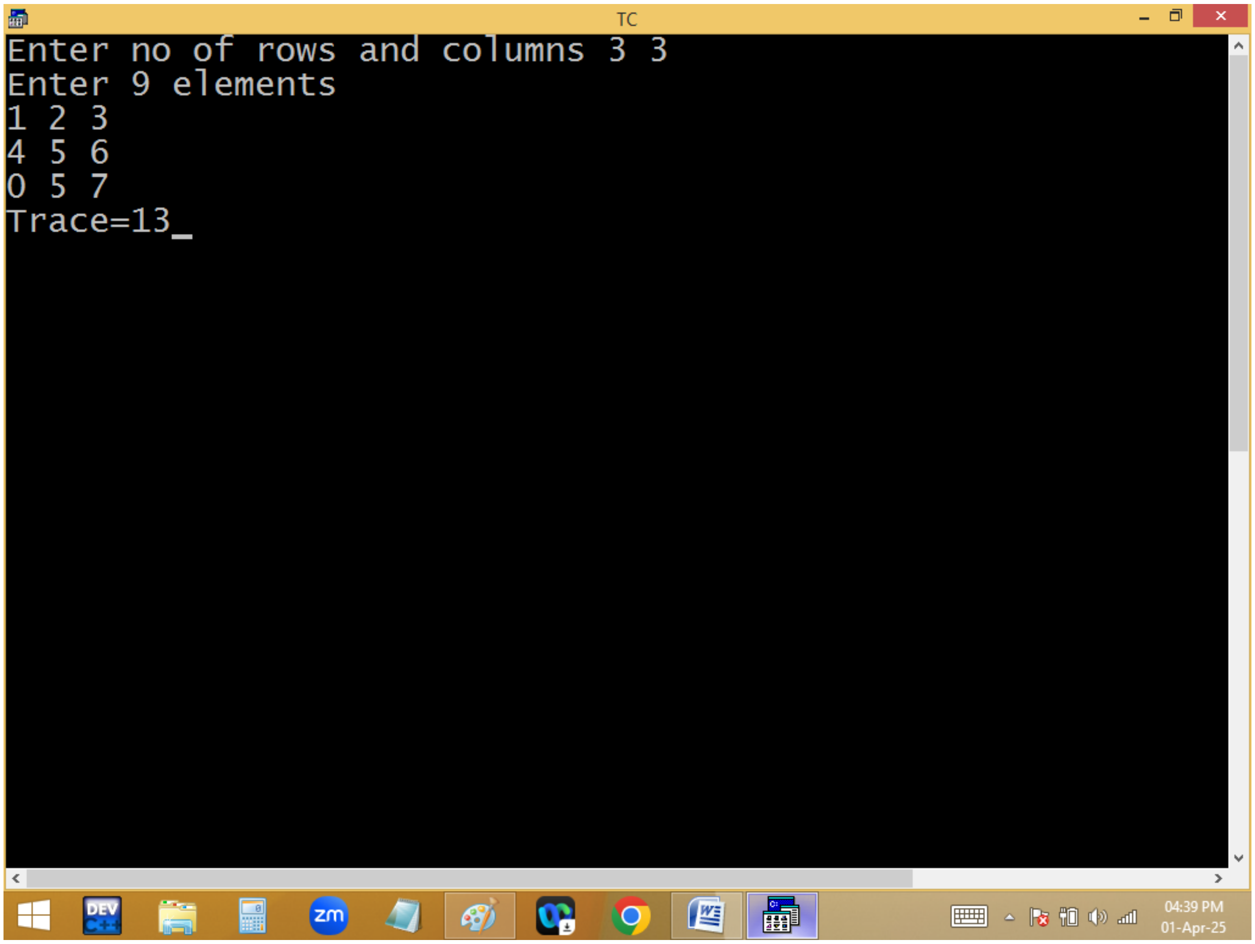
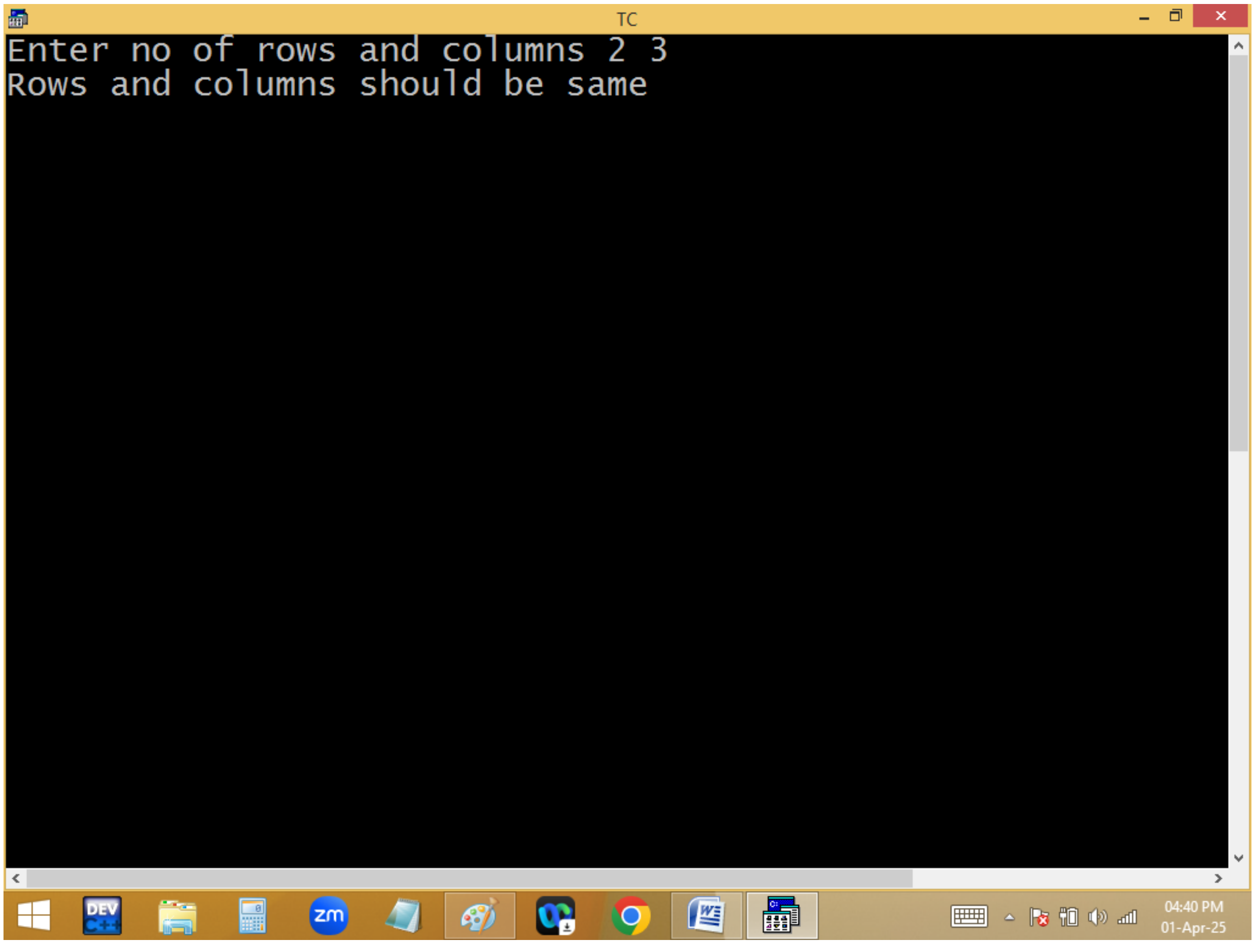
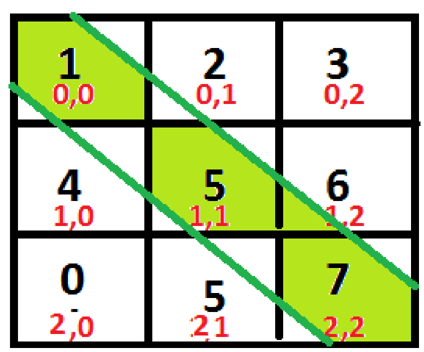

TC
Enter no of rows and columns 3 3
Enter 9 elements
1 2 3
4 5 6
0 5 7
Trace=13_
04:39 PM
01-Apr-25



Sum of principle diagonal elements

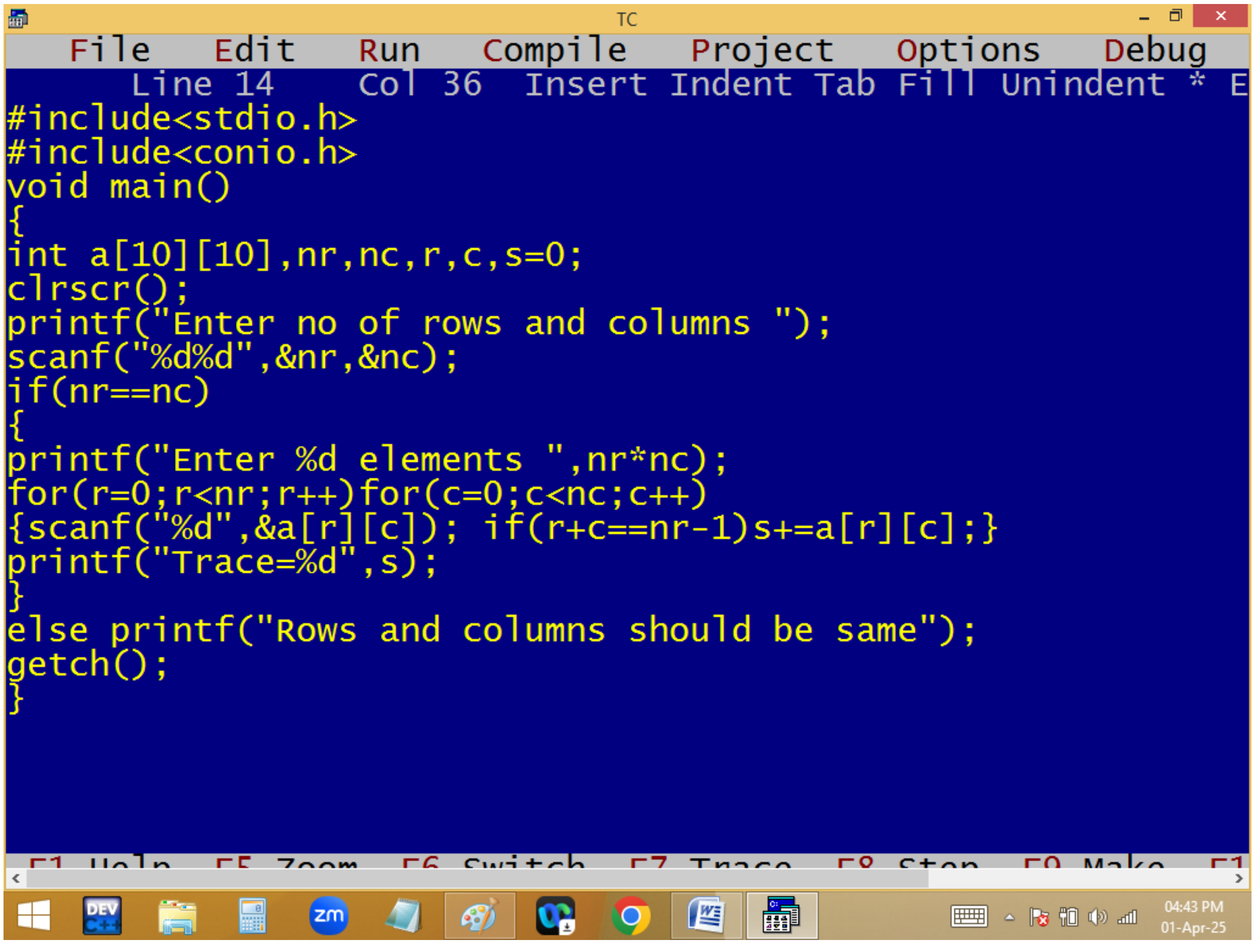
if(r==c) s+=a[r][c];



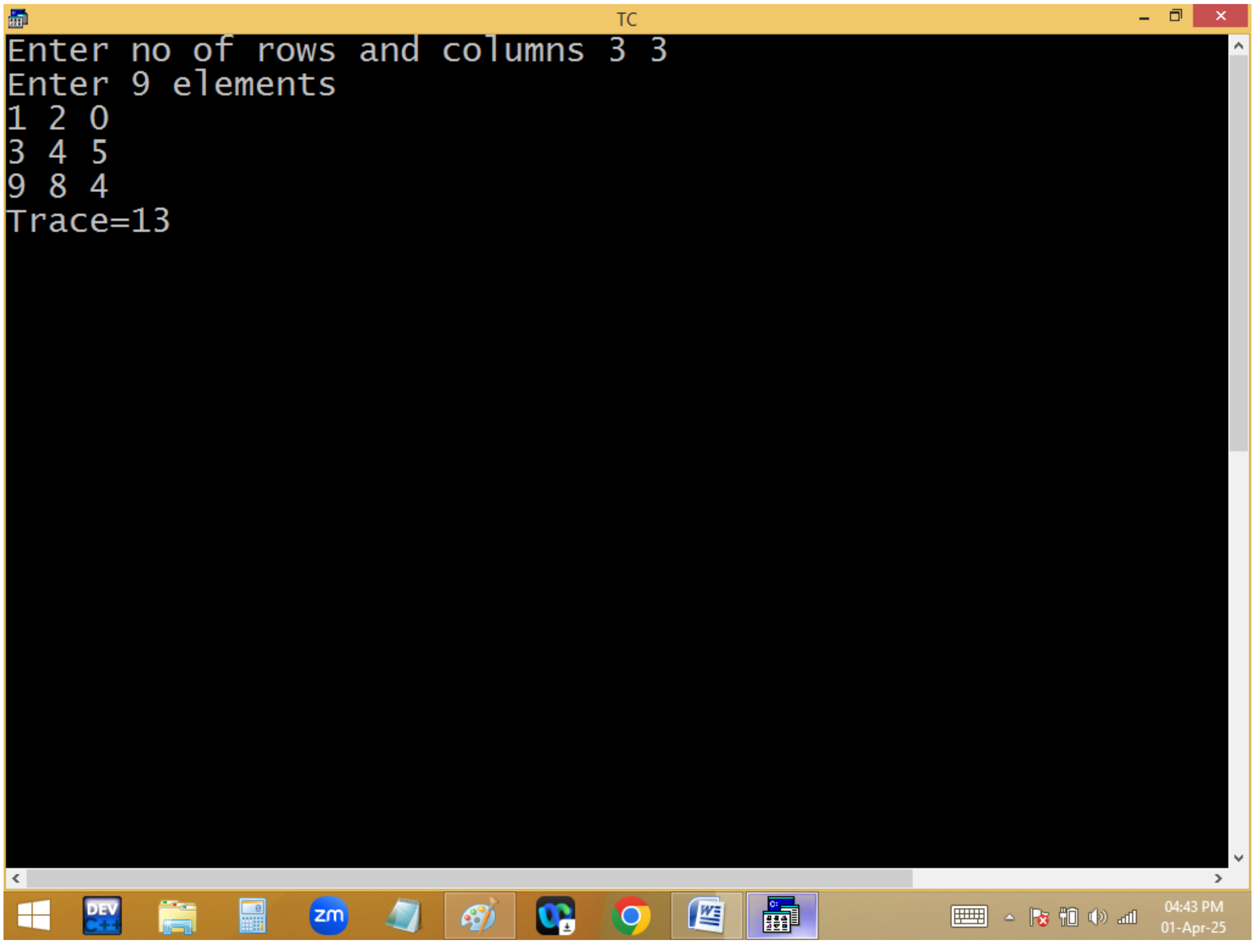
r	c	s
0	0 1 2	0+1+5+7=13
1	0 1 2	
2	0 1 2	

Trace = 13

Finding trace of right diagonal elements:

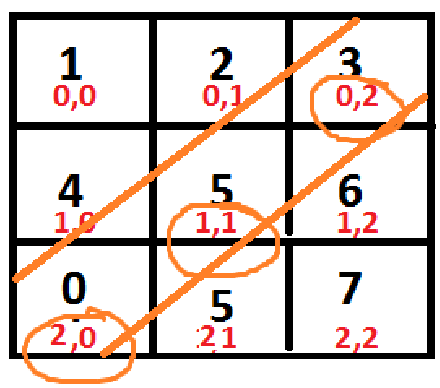



```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,s=0;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
if(nr==nc)
{
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)
{scanf("%d",&a[r][c]); if(r+c==nr-1)s+=a[r][c];}
printf("Trace=%d",s);
}
else printf("Rows and columns should be same");
getch();
}
```



Sum of right diagonal elements: 8

if(r+c == nr-1) s+=a[r][c];



nr

3-1=2

TC

File Edit Run Compile Project Options Debug

Line 12 Col 1 Insert Indent Tab Fill Unindent * E

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\tEven\tOdd\tZero");
puts("-----------------------------------");
for(r=0;r<nr;r++)
{
for(c=e=o=z=0;c<nc;c++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-row\t%d\t%d\t%d\n",r+1,e,o,z);
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Step F9 Make

05:04 PM 01-Apr-25

TC

```
Enter no of rows and columns 3 3
Enter 9 elements
1 2 3
4 5 6
0 5 7
        Even    Odd     Zero
-----------------------------------
1-row   1       2       0
2-row   2       1       0
3-row   0       2       1
```

05:05 PM
01-Apr-25

```
Enter no of rows and columns 4 5
Enter 20 elements
1 2 3 4 0
4 6 8 2 0
1 3 5 7 0
1 3 2 4 5
         Even     Odd      Zero
-----------------------------------
1-row    2        2        1
2-row    4        0        1
3-row    0        4        1
4-row    2        3        0
```

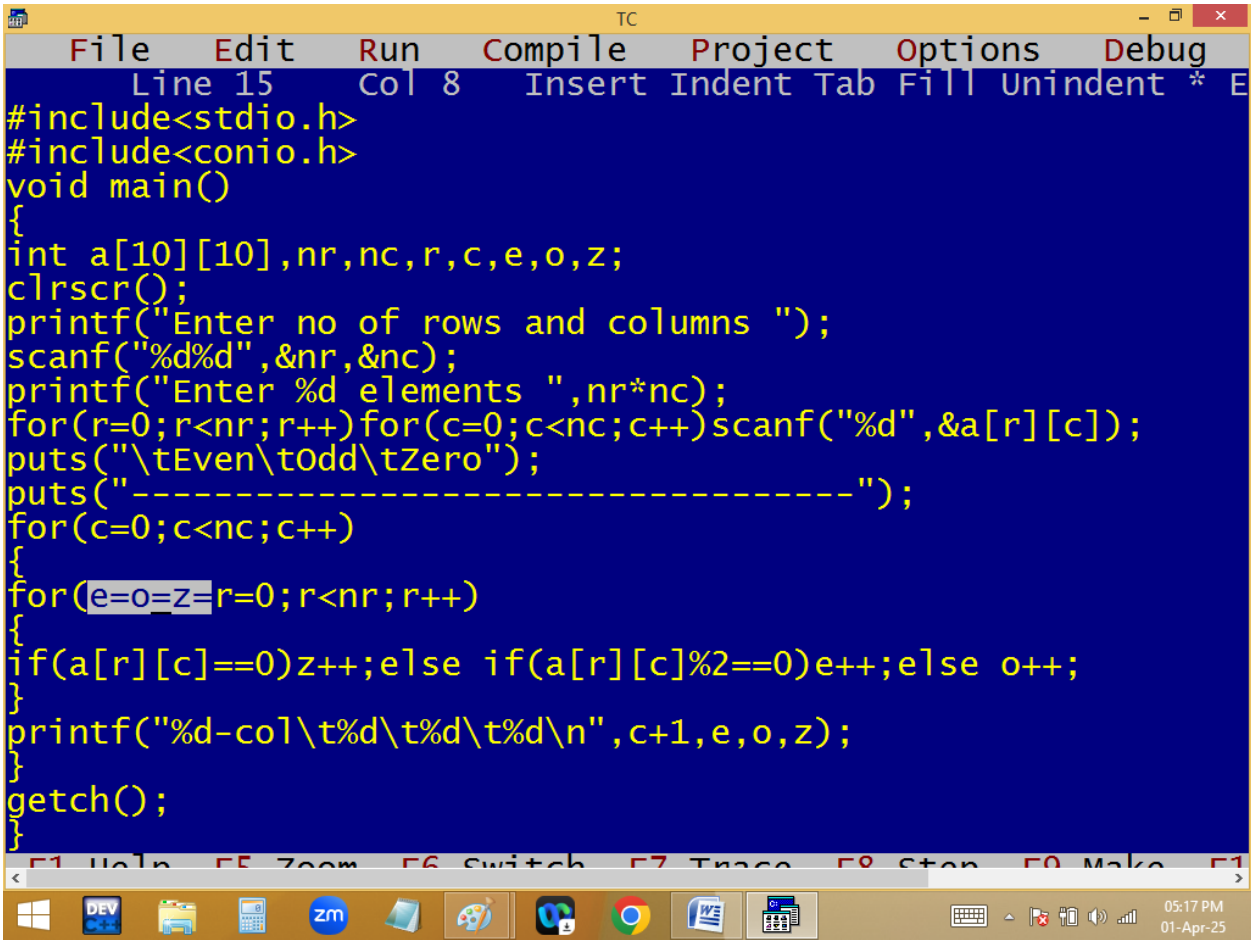
```
p("\t Even\tOdd\tZero");
p("----------------------------------------");

for(r=0;r<nr;r++)
{
for(e=o=z=c=0;c<nc;c++)
{
if(a[r][c]==0)z++;
else if(a[r][c]%2==0)e++;
else o++;
}
p("%d-row\t%d\t%d\t%d\n",r+1,e, o,z);
}
```

r c e o z

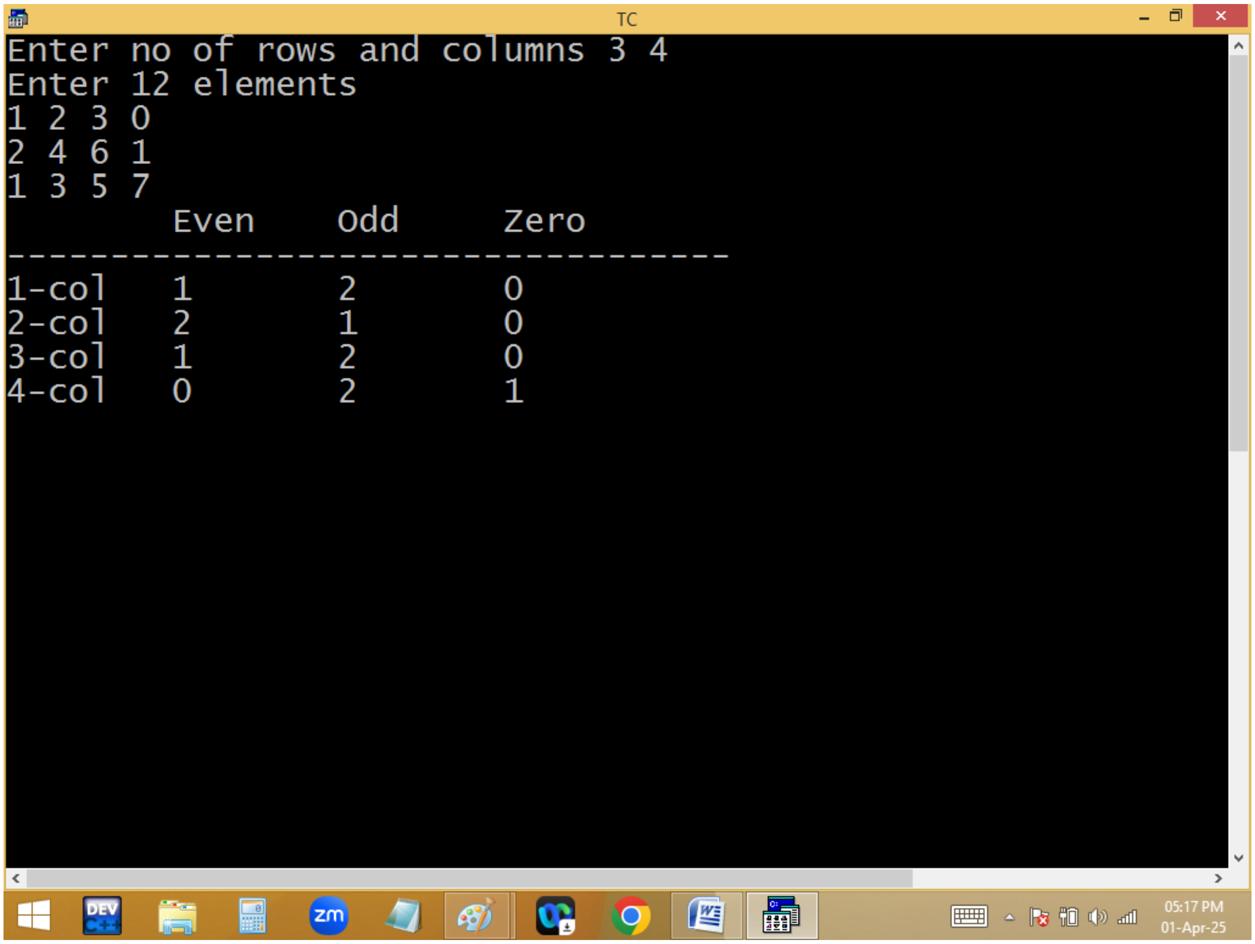
1 (0,0)	2 (0,1)	3 (0,2)
4 (1,0)	5 (1,1)	6 (1,2)
0 (2,0)	5 (2,1)	7 (2,2)

	Even	Odd	Zero
1-row	1	2	0
2-row	2	1	0
3-row	0	2	1

Column wise:



```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\tEven\tOdd\tZero");
puts("-----------------------------------");
for(c=0;c<nc;c++)
{
for(e=o=z=r=0;r<nr;r++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-col\t%d\t%d\t%d\n",c+1,e,o,z);
}
getch();
}
```

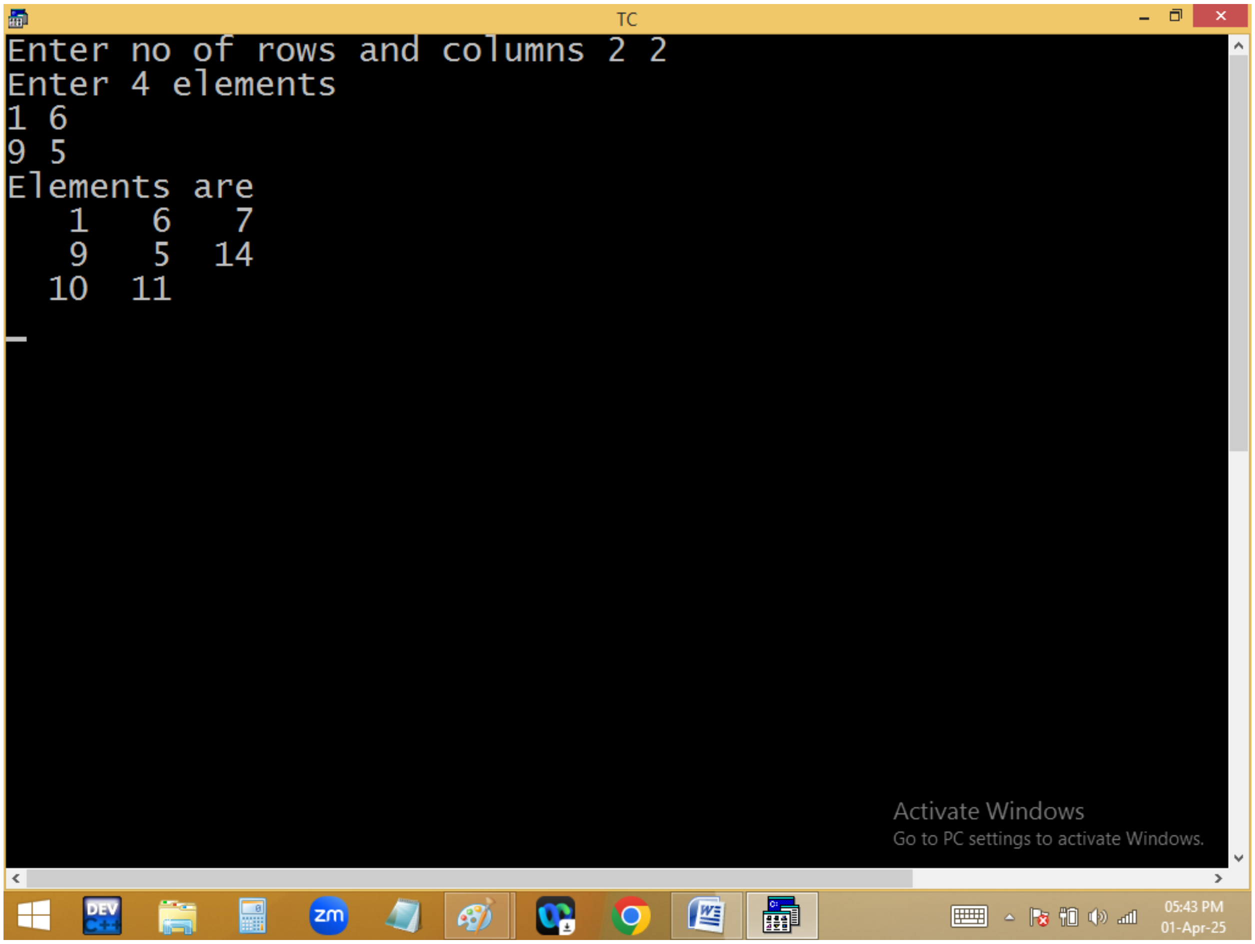


Finding row sum and columns sum:

```
#include<stdio.h>

#include<conio.h>

void main()

{

int a[10][10]={0},nr,nc,r,c,rs,cs;
```

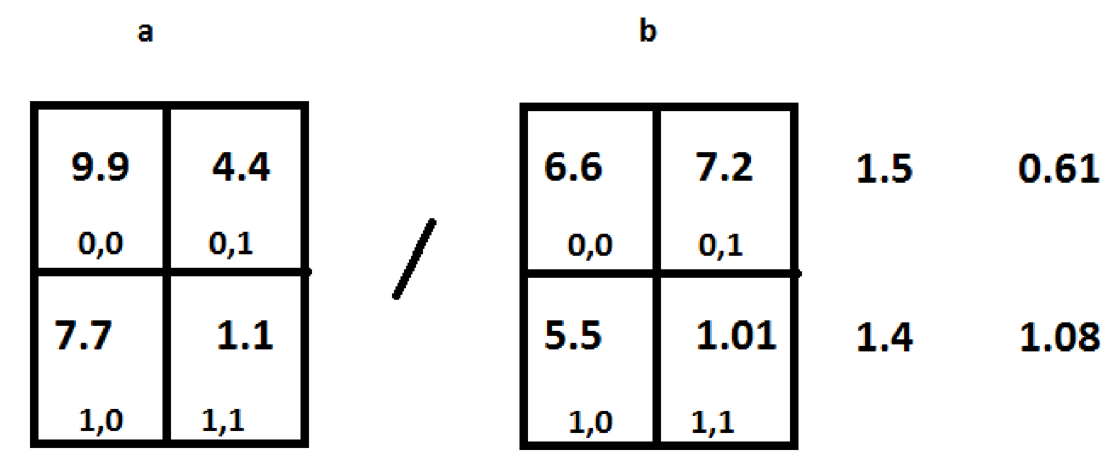
```
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][
c]);
for(r=0;r<nr;r++)
{
for(c=rs=cs=0;c<nc;c++)
{
rs+=a[r][c]; cs+=a[c][r];
}
a[r][c]=rs; a[c][r]=cs;
}
puts("Elements are ");
for(r=0;r<=nr;r++)
```

```
{
for(c=0;c<=nc;c++)
{
 if(r==nr && c==nc)continue; else
printf("%4d",a[r][c]);
}
printf("\n");
}
getch();
}
```



Home work?

Finding the fractions of n*n matrix:



Fractions of n*n matrix

Matrix Multiplication?